

WBSL BRIDGE

Intent-Aware Real-Time Bidirectional Sign Language Communication System
for the Deaf Community of West Bengal

Multimodal Deep Learning · Constrained LLMs · Hybrid Cloud–Edge Inference

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The Problem: A Silent Technological Void

70M+

Deaf & hard-of-hearing individuals rely on sign language worldwide

170

Signs only — the entire static WBSL lexical resource (Wikisigns)

Near 0~

Dedicated AI systems exist for West Bengal Sign Language today

WBSL is linguistically distinct from Delhi ISL and Bangladesh BdSL (Johnson & Johnson, 2016) — yet remains entirely unserved by modern AI, while Google DeepMind's SL2T (Aug 2026) proves consumer-scale ASL translation is possible.

The Intention Problem — most systems translate word-by-word, losing the signer's true intent. A raised eyebrow signals a question; a head shake signals negation. Ignoring these non-manual markers can invert the meaning entirely.

Literature Survey

Progression from sensor-based recognition to national-scale continuous translation

Work	Contribution	Limitation
Sarker & Hoque (2018)	Glove-based BdSL conversion	<i>Hardware-dependent; blind to facial expressions</i>
Islam & Mousumi — Ishara-Lipi	First open-access isolated Bangla sign dataset	<i>Static characters only; no continuous signing</i>
Akash et al. (2023)	Real-time BdSL sentence formation via action recognition	<i>Bangladesh BdSL only</i>
SignISL / Signer.AI	National ISL tools (Indian Railways, IIIT Bangalore)	<i>Standard ISL → Hindi/English only; no WBSL</i>
ISLTranslate / iSign (ACL 2024)	Continuous pan-Indian ISL translation datasets	<i>Output limited to English</i>
Google DeepMind SL2T (Aug 2026)	Real-time ASL-to-English at consumer scale, 100,000+ hrs data	<i>Targets ASL; regional low-resource languages left behind</i>

Identified Research Gaps

G1

Non-Manual Marker Gap

Facial/head grammar cues under-modeled

G2

Regional Linguistic Gap

AI targets standard ISL/BdSL, not WBSL

G3

Data Gap

Only 170 signs; no continuous WBSL dataset

G4

Grammar & Structure Gap

Sign glosses need SOV-aware translation

G5

Resource-Constrained Deployment Gap

Heavy models unusable on CPU-only devices

G6

Real-Time Latency Gap

Low-latency pipeline remains an engineering challenge

G7

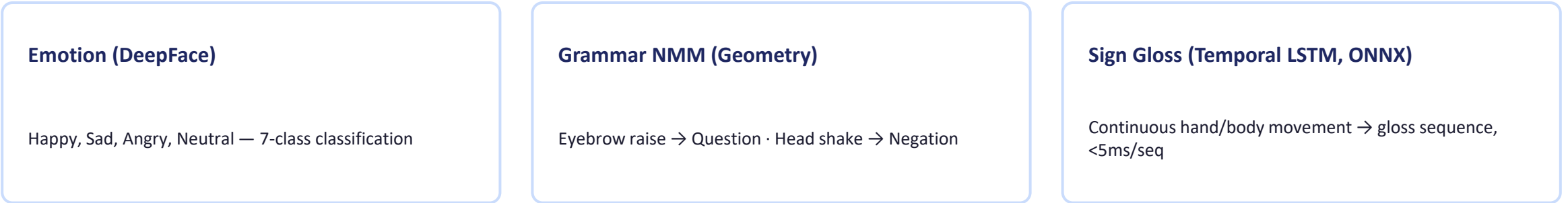
Bidirectional Gap

No dedicated two-way WBSL communication system

Proposed Model — Forward Path (Sign → Bengali)



Three Parallel Modules Feed Intent into the LLM



Bengali Text Generation

Handles SOV word order · Agglutination · Honorifics — gloss + intent reconstructed into grammatically correct Bengali

Proposed Model — Reverse Path (Bengali → Sign)



Key Architectural Innovations

1

Multimodal Intent Capture

Hands + face + body + emotion + grammar markers all feed the LLM

2

User-Selectable LLM Architecture

Cloud OpenRouter or local KoboldCpp, chosen per speed/privacy/internet

3

Privacy by Design

Raw video discarded — only landmarks flow through the pipeline

4

ONNX-Accelerated Inference

LSTM runs at under 5ms per sequence on CPU

LLM Inference Strategy — Measured Benchmarks

User-selectable architecture — no automatic switching; the user controls the trade-off

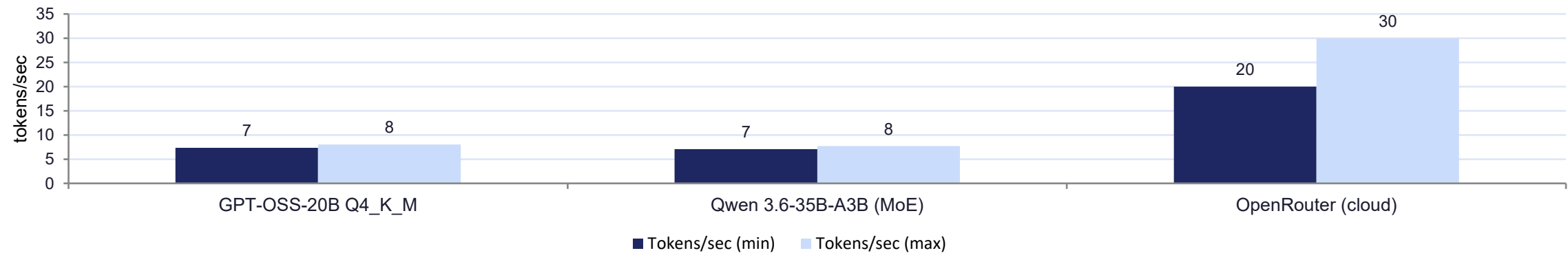
Cloud — OpenRouter API

- Higher generation speed
- Access to powerful cloud models
- Requires internet connectivity
- Speed: ~20–30 tokens/sec

Local — KoboldCpp

- Fully offline after setup
- Greater privacy — data stays on device
- Quantized GGUF models
- Speed: ~7–20 tokens/sec (Ryzen 7U, 16GB)

Benchmark Chart — Local Model Throughput (tokens/sec, Ryzen 7 U, 16GB RAM)



Methodology & Technology Stack

Dataset Strategy

- Primary training: ISLTranslate & iSign (ISL taught in WB schools)
- Baselines: Ishara-Lipi, BdSLW60 for isolated recognition
- Community collection: 50–100 WBSL sentences with Deaf participants (consent-based)
- Supplementary WBSL video review from public platforms for pattern study & annotation, where licensing allows

Technology Stack

Vision: MediaPipe Holistic + OpenCV

Emotion: DeepFace

Temporal Model: PyTorch LSTM + ONNX Runtime

LLM (Cloud): OpenRouter API

LLM (Local): KoboldCpp + GGUF

Reverse Backend: FastAPI + NLP/LLM

Reverse Frontend: React UI

TTS: Piper (offline) / gTTS (online)

Project Timeline

Proposed workflow across the remaining project phase



Evaluation Framework & Expected Results

Evaluation Metrics

- Recognition: Accuracy, Precision, Recall, F1 on held-out signers
- Translation: BLEU, BERTScore, chrF++ for Bengali quality
- Intent Preservation: Deaf signers + hearing raters on 50 test sentences
- Signer Independence: trained/tested on disjoint signers, target $\geq 80\%$ accuracy

Expected Performance

≥30 FPS MediaPipe tracking	<50 ms Emotion detection latency	<5 ms LSTM inference / sequence
1.5–2.5 s End-to-end (cloud) / sign	3–4 s End-to-end (local) / sign	<12 GB Total RAM usage

Future Work

1

Expand WBSL Dataset

Grow community-recorded corpus beyond initial 50–100 sentences for stronger generalization

2

Avatar-Based Sign Generation

Move from clip-mapping to a parametric 3D avatar for smoother reverse-path animation

3

Mobile & Edge Deployment

Optimize the pipeline for smartphones and low-power edge devices

4

Regional Dialect Coverage

Extend beyond WBSL to neighbouring low-resource regional sign variants

5

Field Trials with Deaf Community

Longitudinal usability studies across West Bengal to refine intent accuracy

6

Adaptive LLM Switching

Explore context-aware auto-selection between cloud and local inference

Conclusion

“Can we build a system that understands the intention of a linguistically distinct, data-scarce Deaf community — and speaks back to them in their own language, on their own hardware?”

- Combines MediaPipe, DeepFace, geometric NMM analysis, temporal LSTM, and user-selectable cloud/local LLM inference
- Translates meaning, not just words — preserving question, negation, and emotional intent
- Works offline and online, preserving privacy via landmark-only processing
- Aims to give West Bengal's Deaf community a voice in their own language

Thank You